Terminal functional development

Observations only. This development stage tests a coefficient functional for the saved prime-chain affine witness. It computes no new production LFD or regularity result. The three-family n83 m2–9 fixed-base objective remains active.

Small arithmetic fixtures	n83 synthetic fixtures	Production boundary
12 pairings checked	4 pairings checked	0 new production cases
6 exceptional zeros retained	No actual case binding	LFD fields remain null

terminal-functional-20261030: development counts only; no timing comparison

Native checks. Fourteen release tests passed: six new functional tests and eight inherited donor tests. Twelve small fixtures cross widths 3,5,7 and primes 101,1009 with nonzero and exceptional deltas. Their direct complement-mask pairings equal an independent quadratic-time polynomial-product calculation. All six exceptional targets give zero pairing. A coefficient mutation changes the functional; even widths and zero normalization weights are rejected.

Leading-image controls. All last-block monomial multipliers of degree r−1 were enumerated at each small width, where ell=2r+1. Recorded counts are 1,5,21. Their terminal-top pairings vanish. This enumeration is not a computation of the whole filtered closure. Four synthetic 27-selector fixtures use the ordinary and spike40 weight lists, crossed with target7 and the exceptional target. They are not the42 saved production targets.

Candidate bridge. The proposed functional value is (−2)^r times (sum of weights − 2t), with no second-block factor2 because the terminal intermediate is an unrestricted variable. PRIME_FILTERED_FUNCTIONAL_DESIGN.md gives a proposed syzygy-lifting argument identifying bounded generator products with the filtered closure when all earlier top syzygies are Koszul. Its hypotheses still depend on the conditional Lefschetz argument. These tests do not prove that bridge or establish an LFD bound.

Custody and format. The archive contains three arithmetic JSON files, all native test output, exact sources and Cargo lockfile, and the native test executable. TERMINAL_FUNCTIONAL_FORMAT.json describes the coefficient encoding, variable order, file contents and replay command. The archive is SHA256-addressed in S3; restored verification details are recorded in validation-terminal-functional-development/custody.json.

Next obligations. Validate the complete filtered bridge, add source and base identity joins and replay the actual saved factors, then freeze a new production protocol for the unchanged42case panel. Preserve null LFD fields in older records. No new solving-degree, regularity, timing, ECDLP recovery or elite-base advantage is claimed.

Sources. Caminata–Gorla, Solving degree, last fall degree, and related invariants (arXiv:2112.05579), Sections1–4; PRIME_TERMINAL_DESIGN.md; PRIME_LEFSCHETZ_ARGUMENT.md. Development ID: terminal-functional-20261030.